

Yunxiang Ma

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EDUCATION

Carnegie Mellon University

Master of Science in Computational Design (Applied AI & Machine Learning Track), GPA: 3.86/4.00

Pittsburgh | 2025.09 - 2027.05

Graduate Merit Scholarship

Tsinghua University

Bachelor of Engineering, GPA: 3.73/4.00

Beijing | 2020.09 - 2025.06

National Encouragement Scholarship, 3 Times (Top 20%)

INTERN EXPERIENCES

Bytedance Seed, Global Data

[Seed 2.1](#)

Machine Learning PM Intern | Post-train Data · Data Strategy · Agent Workflow

Beijing | 2026.5 - 2026.8

- **VLM Data Strategy:** Owned data strategy across the evaluation–data–training feedback loop for VLM coding, orchestrating HITL annotation, positive/negative sample construction, synthetic generation, external sourcing, text-data transformation, and diverse execution harnesses to support SFT, RL, and mid-training.
- **3D Coding**
 - **CAD / Floor Plans:** Proposed and delivered an MVP validating the DWG-to-DXF-to-DSL-to-3D code CAD data pipeline. Authored the PRD and independently managed the vendor end to end, and accepted **1.2K+** high-precision human-annotated samples. Partnered with the Engineering team to develop a synthetic-data pipeline, **scaling the dataset to 50K+** samples.
 - **Image-to-Three.js:** To address the scarcity and high production cost of first-person imagery and walkthrough-view data for img2three.js, proposed an alternative solution costing only **0.5%** of a planned data procurement budget. Explored a synthetic-data pipeline leveraging existing three.js code and improved data usability through a custom harness supporting multi-strategy image generation and augmentation.
 - **BenchCAD:** Systematically reverse-engineered the BenchCAD data production methodology and built and optimized the end-to-end pipeline from scratch. Produced **1.2K** expanded-family samples and **11K** observed-family samples, and collaborated with the Data Engineering team to curate **20K+** GitHub samples through keyword-based retrieval and secondary screening, delivering targeted improvements to the BenchCAD benchmark score.
- **Impact:** As PM for VLM 3D Coding, Floor-plan reconstruction featured in the **Seed 2.1 Pro VLM showcase**; contributions supported **Top-10 in Frontend Arena**; surpassing external SOTA models including Gemini 3.1 Pro on in-house benchmark.

Institute of Network Sciences and Cyberspace, Tsinghua University

[arXiv](#) | [Github](#)

Research Assistant | AI Technical Product · XR Development · User Study

Beijing | 2025.6 - 2025.8

- **AI Product Design:** Co-designed **Roomify**, a multimodal generation system that transforms real interior environments through an end-to-end **scene understanding** → **style reasoning** → **content generation** → **spatial registration** pipeline.
- **XR Interaction Design:** Defined key interaction modules for AR creation, including spatial scaffold visualization, object-level editing and feedback-driven correction, improving the controllability and usability of complex GenAI systems.
- **Agent & System Implementation:** Built and integrated an LLM-agent workflow spanning **SLAM3R**, **SpatialLM**, **image generation**, and **3D generation**, supporting spatial-semantic parsing, style-map construction, object-level content generation, and scene reconstruction for automated transformation of physical rooms into themed virtual environments.
- **User Research:** Co-developed evaluation protocols and validated the system with **18 VR users** and **8 design professionals**. Roomify improved perceived immersion by **63% over passthrough** and **26% over fully virtual baselines**; design users reported strong **spatial awareness (5.95/7)** and **creative support (6.08/7)**.

RESEARCH EXPERIENCES

Improving Spatial Depth Reasoning in VLMs through QLoRA Fine-Tuning

[Project](#) | [Github](#)

Group Project | Model Fine-tuning · Benchmark Development · Model Evaluation

Pittsburgh | 2026.01 - 2026.04

- **Data Curation:** Extracted ordered object pairs, instance masks, bounding boxes, and front-back relationships from InstaOrder annotations and COCO images, transforming them into **31.7K direction-balanced, image-disjoint instruction-tuning samples**. Balanced labels through bidirectional sample construction and validated the effects of dataset scale, image isolation, and rendering quality.
- **Experimental Design & Diagnostics:** Designed eight controlled experiments spanning baselines, prompt enhancement, and model fine-tuning. Built a multidimensional evaluation framework covering label bias, occlusion severity, IoU ranges, pairwise consistency and accuracy, and inference quality. The best **7B model achieved 92.65% accuracy and 96.82% consistency**, substantially outperforming the **72B baseline**. Assessed robustness under visual and language prompting conditions and developed a reusable framework for **VLM error analysis and ablation planning**.
- **End-to-End Training:** Co-developed a reproducible **Qwen2.5-VL training and evaluation pipeline**, encompassing spatial prompt design, QLoRA fine-tuning of three 7B models, 7B/72B baseline comparisons, distributed inference on **Modal and NVIDIA B200 GPUs**, model evaluation, and error analysis.

PUBLICATION & HONORS

Roomify: Spatially-Grounded Style Transformation for Immersive Virtual Environments. CHI 2026 Accepted

Demonstrating Roomify: Spatially-Grounded Style Transformation for Immersive Virtual Environments. UIST 2026 Demos Accepted

Selected Honors: CMU NOVA **Most InNOVative Prize**; **Selected** for the Outstanding Works Graduation Exhibition, School of Architecture, Tsinghua University; **First Prize**, Tsinghua University Software Innovation and Creativity Competition; **Second Prize**, 27th Tsinghua University Structural Design Competition; among others.

TECHNICAL SKILLS

Machine Learning & AI: Python, PyTorch, QLoRA, VLM Fine-Tuning & Evaluation, LLM Agents, RAG, SQL, Git, Model Evaluation, Data Analytics, API Integration

Product & Creative Technology: WebXR, User Research, PRD, Figma, Adobe Creative Suite, Rhino, Grasshopper, ComfyUI